# REPORT 4

Repository URL: https://github.com/anakhaskartha/LED.git

## Code Explanation

### 1.1 Header File (led.h)

The header file acts as the public interface, defining structures, constants, and function prototypes:

- **Include Guards:** #ifndef LED_H and #define LED_H prevent multiple inclusions during preprocessing and avoid redefinition conflicts.
- **State Macros:** #define OFF 0 and #define ON 1 establish readable binary state definitions for LED status.
- **Structure Definition:** struct LED encapsulates an integer led_no to identify the LED and an integer pointer *status pointing to its active state value.
- **Function Prototypes:** Lists the names of the functions so the rest of the program knows they exist. ( initialise, set, reset, toggle, display, and deallocate. )

### 1.2 Implementation File (led.c)

This file implements the functions declared in led.h to manage dynamic memory allocation, state mutation, and deallocation:

- initialise(int n):
  - Allocates a contiguous memory block on the heap sized for n struct LED elements using malloc(n*sizeof(struct LED)).

- Iterates from 0 to n - 1 to assign led_no = i + 1, allocates individual memory for each status pointer using malloc(sizeof(int)), and defaults the value to OFF.
- Returns the pointer to the allocated struct LED array.

- set(struct LED *led):
  - Dereferences the status pointer member (*(led->status)) and sets its value to ON.
- reset(struct LED *led):
  - Dereferences the status pointer member (*(led->status)) and sets its value to OFF.
- toggle(struct LED *led):
  - Evaluates *(led->status): if equal to ON, updates it to OFF; otherwise, updates it to ON.
- display(struct LED *leds, int n):
  - Loops across the array and prints fomatted output indicating whether each LED is [LED %d: ON] or [LED %d: OFF] based on the value held at *(leds[i].status).
- deallocate(struct LED *leds, int n):
  - Prevents memory leaks by first freeing each dynamically allocated status pointer inside the loop (free(leds[i].status)).
  - Frees the base array pointer (free(leds)) after all nested allocations have been released.

## 1.3 Driver Program (main.c)

The driver coordinates execution flow and user input:

- **User Input:** Reads the number of LEDs (n) and toggle cycles (m) using scanf.
- **Dynamic Setup:** Calls initialise(n) to construct the array on the heap.

- **Alternating Pattern Configuration:** Iterates through indices 0 to n - 1. Even indices (i % 2 == 0) are assigned reset(&leds[i]), while odd indices are assigned set(&leds[i]).
- **Execution & Display:** Displays the starting state using display(leds, n), then runs an outer loop for m cycles. In each cycle, it toggles every LED via toggle(&leds[i]) and prints the resulting states with display(leds, n).
- **Cleanup:** Calls deallocate(leds, n) to release heap memory before program termination.

## Pushing the Project Files to a GitHub Repository

### Step 1: Create a Repository on GitHub

1. Open your browser and go to github.com.
2. Click the **+** icon in the top-right corner and select **New repository**.
3. Name your repository (e.g., led-toggle-program).
4. Leave it set to **Public** (or **Private**).
5. **Do not** check "Add a README file", .gitignore, or license (keep it completely empty).
6. Click **Create repository**.
7. Copy the repository URL (it looks like [https://github.com/your-username/led-toggle-program.git](https://github.com/your-username/led-toggle-program.git)).

### Step 2: Push the Files from Your VS Code Terminal

In your VS Code terminal (which is already inside your led folder), run the following commands one by one:

**1. Initialize Git in folder:**

```
git init
```

**2. Configure identity :**

```
git config --global user.name "anakhaskartha"
git config --global user.email "anakhaskartha08@gmail.com"
```

**3. Stage all code files:**

git add led.c led.h main.c

**4. Save commit:**

git commit -m "Add LED toggle program"

**5. Set the primary branch to main:**

git branch -M main

**6. Link local folder to GitHub:**

git remote add origin https://github.com/anakhaskartha/LED.git

**7. Push code:**

git push -u origin main